Link Shortening Service Production Architecture and Operating Model

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One Pager Summary Goal Deliver a global link shortening platform that creates compact aliases, redirects users with extremely low latency, and provides analytics and abuse protection. The service must be highly available, horizontally scalable, and cost efficient while supporting enterprise multi tenancy and compliance.

Key Outcomes

Sub fifty millisecond redirect latency from the edge for hot links. Stable create latency and durable storage of mappings. Resilient custom domain support. Actionable analytics for clicks, referrers, and geolocation. Abuse detection including malware and phishing checks, rate limits, and takedown flows.

Primary Assumptions

Peak traffic ranges from tens of thousands to millions of redirects per second globally with bursts. Writes are lower volume than reads with a ratio of one to fifty or higher. Links may be public or tenant scoped with custom domains. Analytics accuracy can be near real time for dashboards and batch complete for deep history.

Functional Requirements

Create short links for long target URLs with optional custom alias, expiration, and metadata. Resolve short codes to targets and issue redirects. Handle tenant scoped namespaces and custom branded domains. Track click events with timestamp, ip to country mapping, user agent, referrer, and campaign parameters. Provide link management APIs for enable, disable, update, and delete. Provide policy and safety features such as blocklists, malware checks, and rate limiting. Offer role based access and audit logs. Export analytics and raw click events to data lake or warehouse.
Non
Availability for redirects at ninety nine point nine nine percent. Availability for create and management at ninety nine point nine five percent. Redirect p95 latency under fifty milliseconds from the edge and under one hundred milliseconds end to end. Create p95 under two hundred milliseconds. Durability with recovery point objective zero for accepted creates. Global cell architecture with regional failover. Compliance with SOC two Type two, GDPR, and CCPA.

Architecture Overview

High level A thin global control plane manages tenants, domains, keys, and configuration. Regional cells run edge redirectors, write and read APIs, caches, and storage. Anycast and geo routing send users to the nearest healthy cell. Redirects are served from edge caches with graceful degradation to regional caches and backing stores.
Creation and Management Path Clients call the management API through API Gateway. The service authenticates via OAuth or OpenID Connect and enforces tenant quotas and rate limits. A write service validates long URLs including format, reachability, and blocklist checks. A unique short code is generated using either preallocated id ranges, a monotonic id generator, or hash with collision resolution. The mapping is written to the durable store with write ahead logging and propagated to caches and edge layers via a message bus.
Redirect Path Users hit an edge endpoint. The edge resolves the code and domain using an in memory key value cache populated by a stream of changes. Cache miss falls back to regional Redis and then to the durable store. After resolution, the redirector records a click event asynchronously and issues an HTTP redirect with strict security headers. The redirector defends against open redirect abuse and validates that the target protocol is allowed.

Analytics Path Edge and regional redirectors emit compact click events to a streaming backbone. Stream processors enrich events with geo mapping, device heuristics, and bot detection, then write near real time aggregates to an analytics store and append raw facts to the lake for long term queries. Dashboards query

aggregates for fast time to glass while analysts can query the lake for deep history.
Technology Stack Edge and auth Cloud CDN, serverless edge workers or compute at edge, global load balancer, API Gateway, WAF, OAuth two point one or OpenID Connect, mTLS in the mesh. Streaming Kafka with Schema Registry and Kafka Connect or a managed equivalent. Processing Apache Flink for enrichment and aggregation. Caches Redis Cluster for regional hot sets and in process caches at the edge. Durable store DynamoDB or Scylla or Cassandra for mappings and metadata with tenant id and code keys. Object storage S3 or GCS for raw events in Parquet with Iceberg or Delta. Analytics ClickHouse or BigQuery or Snowflake for aggregates and dashboards. Platform Kubernetes for services with autoscaling. Terraform or Pulumi and Helm for infrastructure as code. Argo CD and Argo Rollouts for progressive delivery.

Storage Strategy

Hot path Edge in memory maps keyed by domain and code with compact structs.
Regional Redis Cluster holds the hot working set with short time to live and write through on miss. Redis memory is protected with eviction policies and per key size limits. Durable path uses DynamoDB or Scylla with composite key tenant and code and a global secondary index by domain and code to support custom domains. Strongly consistent writes are used for create and management. Event path uses Kafka for resilient ingestion and fanout. Cold path places raw click events in Parquet with hourly partitions in object storage. Aggregates are written to ClickHouse or an equivalent analytical database for low latency dashboards. Indexing is limited to essential keys to avoid write amplification.
Scaling Approach Cells operate per region with their own caches and stores. Codes are generated to avoid cross region coordination by using id ranges or base n encodings that include cell prefix and shard. Edge caches subscribe to change streams with sequence tokens to apply updates. Redis shards scale horizontally with consistent hashing and hash tags for domain or code collocation. Durable store capacity scales with adaptive partitions and on demand throughput. Analytics processing scales by partitioning Kafka topics by tenant and domain and time. CDN capacity scales independently. Back pressure is managed by the streaming backbone and edge gracefully degrades to regional caches when the durable store is impaired.
Latency Budgets and Service Levels Edge lookup and redirect under ten milliseconds in cache hit cases. Regional Redis miss retrieval under five milliseconds. Durable store lookup under twenty milliseconds p95. Create path validation and write under two hundred milliseconds p95. Ingest to aggregate visibility under one minute p95. Redirect availability at ninety nine point nine nine percent. Create and management availability at ninety nine point nine five percent.

Data Models

Link tenant id, domain, code, target url, created at, created by, expires at, enabled flag, tags, campaign, attributes json, last updated at, version. Domain tenant id, domain name, dns status, certificate id, cname target, created at, status, last updated at. ClickEvent tenant id, domain, code, timestamp, ip prefix, country, region, device family, user agent hash, referrer source, utm fields, bot score, source edge id. Aggregate tenant id, domain, code, bucket start, granularity, count, unique ip estimate, country distribution sketch, referrer breakdown sketch. AuditLog actor id, tenant id, action, target, timestamp, diff preview. All records include trace identifiers for correlation and optional customer managed encryption key references.

Key Algorithms and Workflows

Code generation Option A uses a cluster friendly id generator issuing
monotonically increasing ids per cell with randomization to avoid inference. Option B uses hash of target url with collision resolution by suffix. Option C supports custom aliases with conflict checks. Base sixty two or base fifty eight encoding yields short strings. Vanity and reserved prefixes are enforced by policy. Edge cache fill A change log of link mutations is published to a stream. Edge workers maintain a tailer that

applies updates in order with checkpoints. New edges bootstrap using a snapshot plus change replay.

Abuse and safety Targets are validated via allow list of schemes and optional real time security checks. Bot detection based on heuristics and third party signals downweights analytics. Rate limits per client, per tenant, and per code prevent brute force enumeration. Sensitive targets such as admin panels are blocked by policy. Analytics accuracy Unique counts use HyperLogLog sketches per bucket. Country or referrer distributions use mergeable sketches. Sampling can be applied for extreme codes and corrected in aggregates. Backfills repair late arriving events. Consistency Read after write consistency for creates within tenant is achieved by writing to durable store and synchronously priming regional Redis and publishing an edge update. Redirects remain available using last known mapping even if management path is degraded.

Security Reliability and Compliance Encryption in transit with TLS one point three and at rest with KMS keys. Optional per tenant customer managed keys. Strict input validation and canonicalization of URLs. OAuth scopes and RBAC for management APIs and domain operations. Signed webhooks for outbound events. WAF and bot mitigation at edge. Audit logs for all link and domain changes. Privacy controls for data subject requests and retention policies for event data. Multi AZ within each region and regional isolation across cells. Disaster recovery with tested backups of mappings and object storage versioning.

Multi Tenancy Tenant id propagates across all keys and topics. Logical isolation by tenant scoped partitions, caches, and capacity budgets. Optional hard isolation by dedicated cells and virtual private clouds for large customers. Per tenant rate limits, quotas, and dashboards. Per tenant encryption keys and custom domains with automated certificate issuance.

CI and CD and Infrastructure as Code All infrastructure defined with Terraform or Pulumi and Helm for Kubernetes services. Progressive delivery with Argo Rollouts using canary steps gated on redirect p95 latency, error rate, and cache hit ratio. Automatic rollback when service level objectives burn. Schema registry and contract tests for streams and APIs. Database migrations use online techniques and dual write or shadow read windows where necessary.

APIs and Integration Points Public REST endpoints include create link, update, delete, fetch, list by tenant and by domain, and bulk import. Redirect endpoint resolves domain and code and issues a redirect. Webhooks notify on link creation, update, delete, and disabled events. Analytics endpoints provide time series and breakdowns. Admin endpoints manage tenants, domains, certificates, and policies. SDKs and examples are provided for common languages. Observability and Operational Practices OpenTelemetry for traces and metrics. Dashboards show redirect tail latency, cache hit rate, errors by class, create latency, stream lag, durable store latency, and aggregate freshness. Alerts wired to error budget policies. SLO aware autoscaling based on p95 tail, Redis ops, and stream lag. Runbooks cover cold cache bootstrap, Redis shard failover, durable store throttling, domain certificate renewal, and stream replays.

Capacity Planning and Cost Estimate A busy region serving two hundred thousand redirects per second and ten thousand creates per second may use a Redis cluster with twelve to twenty four shards, a DynamoDB or Scylla cluster sized for read heavy workloads, and Kafka with six to nine brokers. Edge capacity is dominated by CDN and serverless workers. Analytics storage costs scale with event volume and retention. Rough monthly cost order of magnitude ranges from twenty thousand to eighty thousand per region depending on traffic, retention, and cache sizes.

Testing Strategy and Failure Drills

Unit tests for code generation, parsing, validation, and policy enforcement. Property tests to ensure id uniqueness and collision handling. Contract tests for APIs and stream schemas. Load tests for redirect hot paths with cache miss scenarios and for create bursts. Soak tests for cache churn and edge update replay. Chaos drills for Redis shard loss, durable store throttling, stream backlog, certificate failures, and regional isolation. Disaster recovery drills validate restore of mappings and event reprocessing.

Phased Build Plan

MVP three months single region cell with create and redirect, Redis plus DynamoDB mapping store, Kafka for change log and analytics, minimal dashboards, and basic policy checks. Phase two six to nine months multi region cells with edge cache tailers, custom domains and automated

certificates, analytics aggregates and export, and canary release process. Phase three twelve months advanced abuse detection, tenant hard isolation options, higher scale code generation, cost optimization with tiered caches, and enterprise features including SSO and customer managed keys.
Key Choices Trade Offs Risks and Mitigations Edge heavy caching yields very low latency but requires robust change streams and snapshot mechanisms and can serve stale mappings briefly. Durable store selection trades operational ease and cost. Hash codes are deterministic and compact but require collision handling while numeric id paths are simple but predictable without randomization. Analytics sketches reduce cost and improve speed but introduce small error which is acceptable for dashboards. Multi region isolation keeps blast radius small but requires duplicated capacity and a control plane to coordinate configuration across cells.